microscopic scales, the average speed of the water molecules decreases, but that's about all. When you decrease the temperature to 0 degrees Celsius, however, you suddenly see that something drastic occurs. The liquid water begins to freeze and turn into solid ice. As discussed in the preceding chapter, this is a simple example of a phase transition. For our present purpose, the important thing to note is that the phase transition results in a decrease in the amount of symmetry displayed by the H2O molecules. Whereas liquid water looks the same regardless of the angle from which it is viewed—it appears to be rotationally symmetric—solid ice is different. It has a crystalline block structure, which means that if you examine it with adequate precision, it will, like any crystal, look different from different angles. The phase transition has resulted in a decrease in the amount of rotational symmetry that is manifest.

Although we have discussed only one familiar example, the point is true more generally: as we lower the temperature of many physical systems, at some point they undergo a phase transition that typically results in a decrease or a "breaking" of some of their previous symmetries. In fact, a system can go through a series of phase transitions if its temperature is varied over a wide enough range. Water, again, provides a simple example. If we start with H2O above 100 degrees Celsius, it is a gas: steam. In this form, the system has even more symmetry than in the liquid phase since now the individual H2O molecules have been liberated from their congested, stuck-together liquid form. Instead, they all zip around the container on completely equal footing, without forming any clumps or "cliques" in which groups of molecules single each other out for a close association at the expense of others. Molecular democracy prevails at high enough temperatures. As we lower the temperature below 100 degrees, of course, water droplets do form as we pass through a gas-liquid phase transition, and the symmetry is reduced. Continuing on to yet lower temperatures, nothing too dramatic happens until we pass through 0 degrees Celsius, when, as above, the liquid-water/solid-ice phase transition results in another abrupt decrease in symmetry.

Physicists believe that between the Planck time and a hundredth of a second ATB, the universe behaved in a very similar way, passing through at least two analogous phase transitions. At temperatures